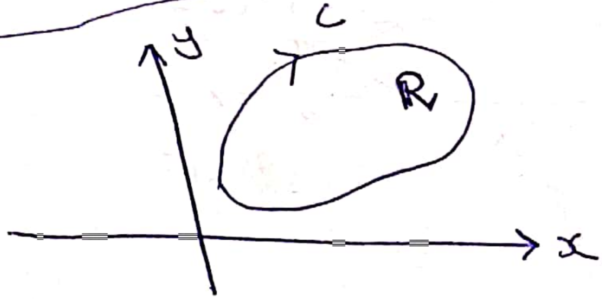
Green's theorem (in the plane)

(Transformation b/w Double and line integral)

Let R be a closed bounded region in the xy-plane whose boundary C consist of finitely many smooth curves. Let $P(x,y)$ and $Q(x,y)$ be function that are continuous, then

$$\boxed{\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \oint_C P\,dx + Q\,dy} \quad \text{—(1)}$$

(Double integral) (line integral)



$$\boxed{\vec{F} = P(x,y)\hat{i} + Q(x,y)\hat{j}} \quad \text{—(2)}$$

Ex (verify Green's theorem)

Let $\vec{F} = (y^2 - 7y)\hat{i} + (2xy + 2x)\hat{j}$ —(1) , $C: x^2 + y^2 = 1$ —(4)

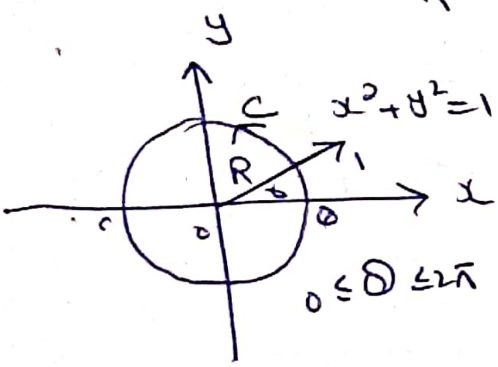
$P = y^2 - 7y$ —(2)

$Q = 2xy + 2x$ —(3)

Sol (Proof) L.H.S = R.H.S (To show)

01

02 $$\text{L.H.S} = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \quad \text{—(4)}$$



Double integral (in polar coordinate)

$dA = r\,dr\,d\theta$ —(5)

$x = r\cos\theta$ —(6)

$y = r\sin\theta$ —(7)

03 $\frac{\partial Q}{\partial x} = 2y + 2$ —(8) , $\frac{\partial P}{\partial y} = 2y - 7$ —(9)

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \cancel{2y} + 2 - \cancel{2y} + 7 = 9 \quad \text{—(10)}$$

(9) in (4),

(4) $\Rightarrow$ $L.H.S = \iint_R (9)\, dA$ — (11)

$$= \int_0^{2\pi}\int_0^1 9\, r\, dr\, d\theta \quad — (12)$$

$$= 9\int_0^{2\pi} \left[\frac{r^2}{2}\right]_{r=0}^{r=1} d\theta$$

$$= \frac{9}{2}\int_0^{2\pi} d\theta = \frac{9}{\cancel{2}} \cancel{2}\pi = 9\pi \quad — (13)$$

Part 02

$R.H.S = \oint_C P\,dx + Q\,dy$ — (14)

$\vec{r}(t) = [\cos t, \sin t] = \cos t\, \hat{i} + \sin t\, \hat{j}$ — (15)

$x(t) = \cos t$, (16) $\quad y(t) = \sin t$ (17)

$dx = -\sin t\, dt$, (18) $\quad dy = \cos t\, dt$ (19)

$P(x(t), y(t)) = \sin^2 t - 7\sin t$, (20) $\quad Q(x(t), y(t)) = 2\cos t \sin t + 2\cos t$ (21)

put (15) to (21) in (14)

$$(14) \Rightarrow R.H.S = \int_0^{2\pi} \Big[(\sin^2 t - 7\sin t)(-\sin t\, dt) + (2\cos t \sin t + 2\cos t)(\cos t\, dt) \Big] dt \quad -(22)$$

$$= \int_0^{2\pi} \left(\cancel{-\sin^3 t}^{\,0} + \cancel{7\sin^2 t}^{\,\pi} + \cancel{2\cos^2 t \sin t}^{\,0} + \cancel{2\cos^2 t}^{\,2\pi} \right) dt \quad -(23)$$

$$= 9\pi \quad -(24)$$

(Part 3)

so, from (13) & (24)

$$\Rightarrow \boxed{19\pi = 9\pi}$$

<u>Hint</u>

01 $\int \sin^3 t \, dt = \int \sin^2 t \sin t \, dt = \int (1-\cos^2 t) \sin t \, dt$

Let $u = \cos t$ $du = -\sin t \, dt$

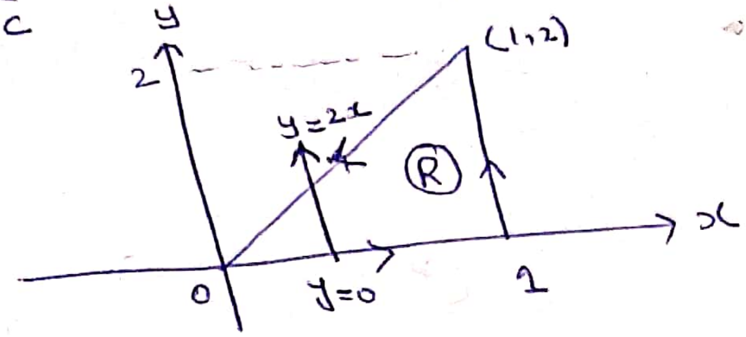
02 $\int \cos^2 \sin t \, dt = -\int \cos^2 \sin t \, dt = -\dfrac{\cos^3 t}{3}$

03 $\sin^2 t = \dfrac{1-\cos 2t}{2}$, $\cos^2 t = \dfrac{1+\cos 2t}{2}$

Ex

Use Green's theorem to evaluate

$$\oint_C xy\,dx + x^2y^3\,dy \quad , \text{ where } C$$



$0 \le x \le 1 \quad , \quad 0 \le y \le 2x$

Sol

01

$$\int_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dy\,dx \quad \text{— (01)}$$

02

$$P = xy \quad \text{— (02)} \quad , \quad Q(x,y) = x^2y^3 \quad \text{— (03)}$$

$$\frac{\partial Q}{\partial x} = 2xy^3 \quad \text{— (04)} \quad , \quad \frac{\partial P}{\partial y} = x \quad \text{— (05)}$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2xy^3 - x \quad \text{— (06)}$$

03

$$\int_C P\,dx + Q\,dy = \int_{x=0}^{x=1}\int_{y=0}^{2x} (2xy^3 - x)\,dy\,dx \quad \text{— (07)}$$

$$= \int_0^1 \left(2x \cdot \frac{y^4}{\cancel{4}_{2}} - xy \right) \Bigg|_{y=0}^{y=2x} dx$$

$$= \int_0^1 \left(2x\left((2x)^4 - 0\right) - x\left(2x - 0\right) \right) dx$$

$$= \int_0^1 \left(8x^5 - 2x^2\right) dx$$

$$= \frac{2}{3}.$$

Exercise

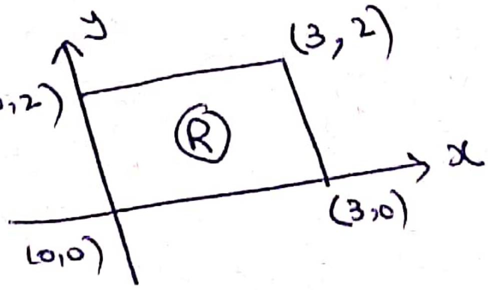
Verify Green's theorem

01 Let $F(x,y) = (x-y)\hat{i} + x\hat{j}$

and R is bounded by unit circle $C: x^2 + y^2 = 1$.

02 $$\oint_C \left(\frac{1}{2}xy^4\right)dx + \left(\frac{1}{2}x^4 y\right)dy$$

along C:



$0 \le x \le 3, \quad 0 \le y \le 2$